



Basic Sales Summary with SQLite and Python



Objective

It demonstrates how to:

- Create and connect to a SQLite database.
 - Create a simple sales table.
 - Insert sample sales data.
 - Query total quantity and revenue per product using SQL.
 - Visualize the revenue summary using a bar chart in matplotlib.
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Tools & Libraries

- **SQLite** (sqlite3): Lightweight built-in SQL database in Python.
 - **Pandas** (pandas): For loading and manipulating SQL query results.
 - **Matplotlib** (matplotlib.pyplot): To generate basic bar chart visualizations.
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Dataset

A small, hardcoded dataset is used to populate the sales table with entries including:

- product (e.g., Apples, Bananas, Oranges)
 - quantity sold
 - price per unit
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SQL Query Used

```
SELECT
    product,
    SUM(quantity) AS total_quantity,
    ROUND(SUM(quantity * price), 2) AS total_revenue
FROM sales GROUP BY product
```

This aggregates data to show:

- Total quantity sold per product.
- Total revenue per product (quantity × price), rounded to 2 decimal places.



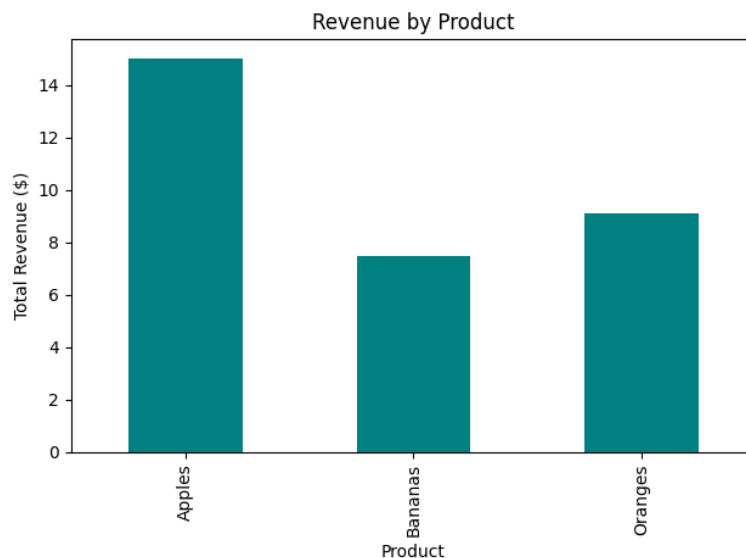
Output

- **Console:** Displays a table with total quantity and revenue per product.

```
Sales Summary:
```

	product	total_quantity	total_revenue
0	Apples	30	15.0
1	Bananas	25	7.5
2	Oranges	13	9.1

- **Plot:** Bar chart of total revenue by product is shown using matplotlib.



- **Image File:** Chart is saved as sales_summary_chart.png.

```
# Plot revenue as a bar chart
df.plot(kind='bar', x='product', y='total_revenue', color='teal', legend=False)
plt.title("Revenue by Product")
plt.xlabel("Product")
plt.ylabel("Total Revenue ($)")
plt.tight_layout()
plt.savefig("sales_summary_chart.png") # Optional: save as image
plt.show()
```

Cleanup

- Database connection is properly closed at the end to free resources.

```
# Close connection  
conn.close()
```

Conclusion & What I Learned

By completing this task, I learned how to:

- Set up and work with a SQLite database using Python.
- Create and populate tables programmatically.
- Write and execute basic SQL queries from within Python.
- Use pandas to manipulate SQL query results for easy data analysis.
- Generate clear visualizations using matplotlib to represent SQL data insights.

This task provided a hands-on introduction to combining SQL with Python for simple but effective data analysis, reinforcing both technical skills and visualization techniques.